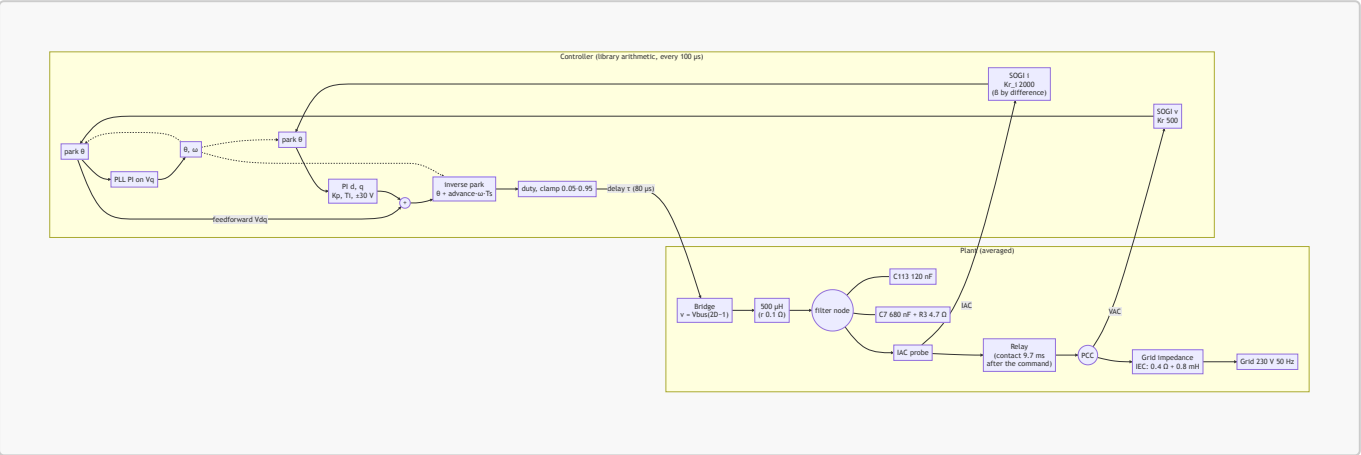
D-GFL-01: connection fidelity ladder, IEC grid

Design test (ADR 0004): averaged model, INFO only, no verdict. Candidate C (Kp 1 V/A, Ti 5 ms, current SOGI 2000 rad/s, advance 0.5 period), current PI active from the relay command unless stated. Grid: 0.4 Ω, 0.8 mH. Generated by `tests/l1/design/ladder.py --report`.

Plant and loop

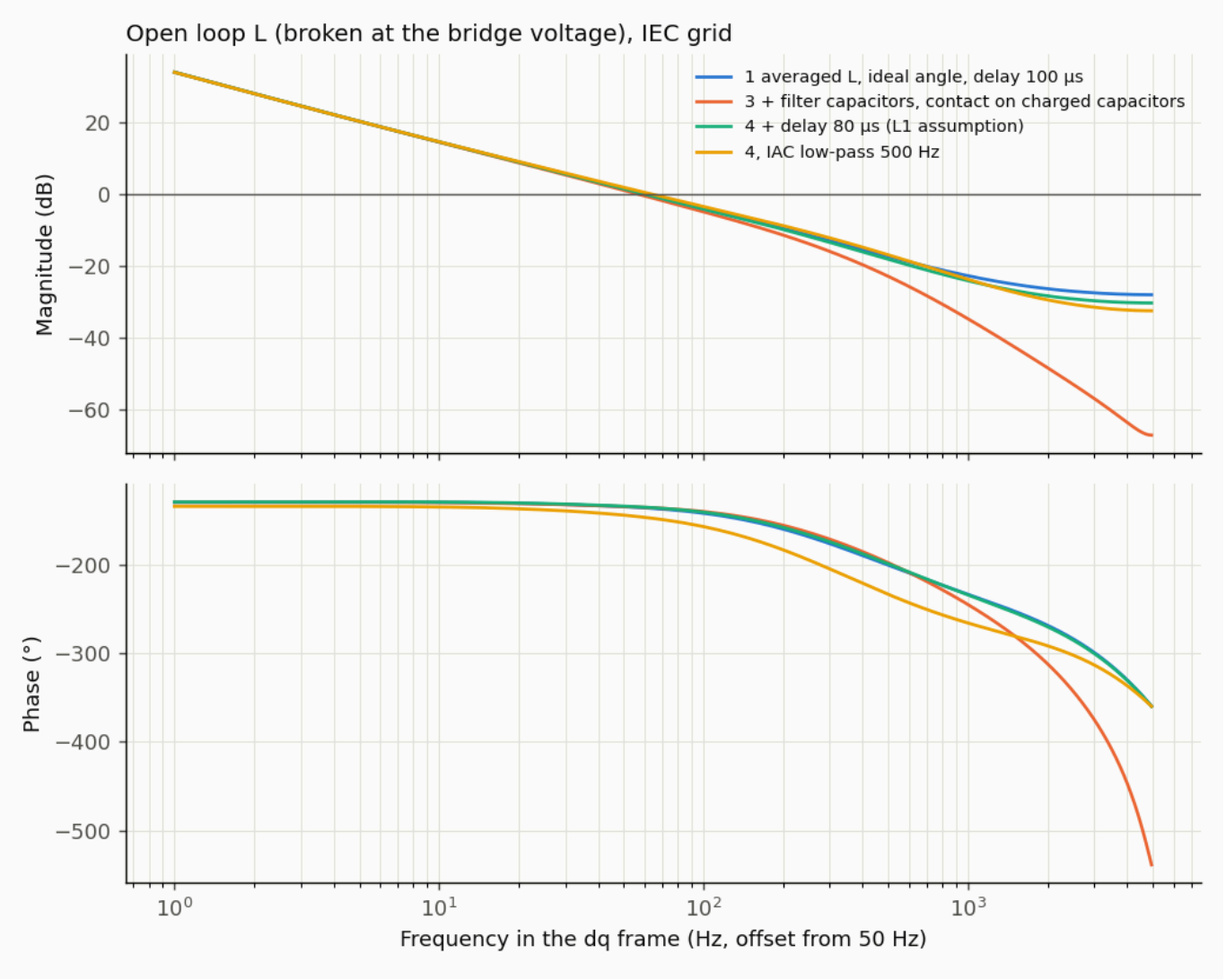


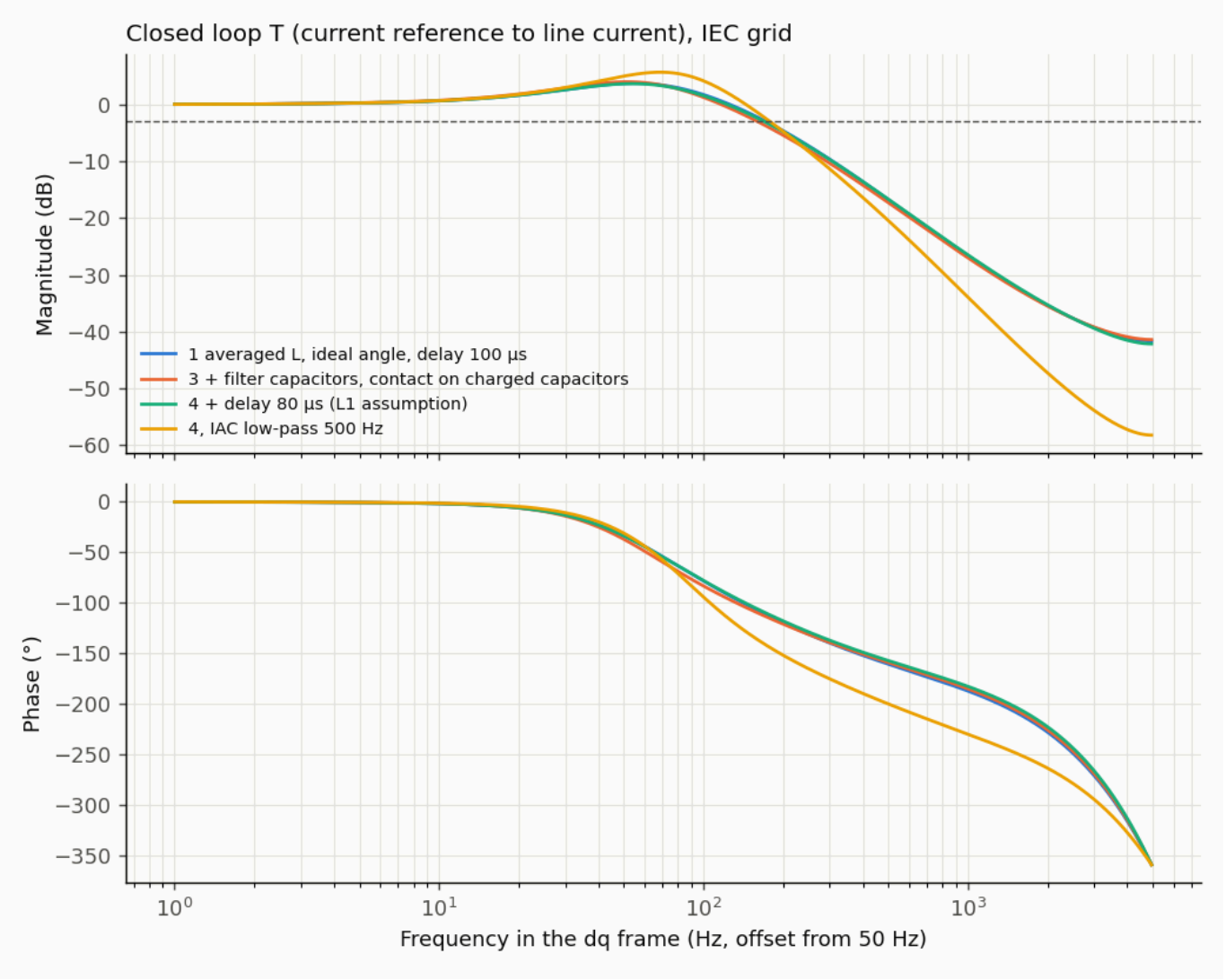
Margins and bandwidth (linear model, relay closed, ideal angle)

Margins are open-loop properties; the closed loop is described by its bandwidth and its resonance peak. Frequencies in the dq frame. Rung 2 (PLL) has no linear row: the linear model uses the ideal angle.

Case	Open loop: crossover (Hz)	Open loop: phase margin (°)	Open loop: gain margin (dB)	Closed loop: bandwidth −3 dB (Hz)	Closed loop: peak (dB)	Closed loop: peak at (Hz)
1 averaged L, ideal angle, delay 100 µs	60.6	44.5	13.7	172	3.7	55
3 + filter capacitors, contact on charged capacitors	56.6	45.5	18.3	160	3.97	51.4
4 + delay 80 µs (L1 assumption)	59.9	45.3	14.6	170	3.64	53.8
4, IAC low-pass 500 Hz	63.7	32.9	8.27	178	5.66	68.6
4, no output advance	60	43.8	13.5	173	3.92	55.5

Case	Open loop: crossover (Hz)	Open loop: phase margin (°)	Open loop: gain margin (dB)	Closed loop: bandwidth −3 dB (Hz)	Closed loop: peak (dB)	Closed loop: peak at (Hz)
4, Kp 0.5	30.4	31.5	12.4	82.7	5.08	32.6





Connection transient (time domain, trip disabled)

Contact on a voltage zero crossing; the 1.5 A bench trip is reported, not applied, so the whole transient is visible.

Case	Peak line current, first 10 ms (A)	Would trip at 1.5 A	Line current, first 5 ms (A)	|I_dq| seen by the controller, first 5 ms (A)	Transient above 0.15 A (ms)	I_d after the 0.1 A step (A)
1 averaged L, ideal angle, delay 100 µs	1.29	no	1.29	2.41	21.8	0.1
2 + library PLL	3.32	yes	1.26	3.54	98.2	0.101
3 + filter capacitors, contact on charged capacitors	11.7	yes	4.36	11.7	159	0.1

Case	Peak line current, first 10 ms (A)	Would trip at 1.5 A	Line current, first 5 ms (A)	|I_dq| seen by the controller, first 5 ms (A)	Transient above 0.15 A (ms)	I_d after the 0.1 A step (A)
4 + delay 80 µs (L1 assumption)	1.65	yes	1.33	2.6	81.9	0.1
4, PI from the feedback (library today)	7.73	yes	3.84	3.97	93	0.0998
4, IAC low-pass 500 Hz	2.02	yes	1.53	3.32	80.9	0.1
4, no output advance	2.61	yes	2.61	6.35	91.9	0.1
4, Kp 0.5	4.01	yes	1.93	3.5	190	2